Highlights

Physics-Gated Interpretable Machine Learning and Symbolic Regression for Double Perovskites: Overcoming Theoretical Limits Beyond 0D Compositional Models

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- Physics-gated domain routers resolve zero-inflation in 0D double perovskite ML.
- Harrison tight-binding and tie-line engines elevate 0D models beyond SOTA limits.
- Achieves 109.62% of theoretical limit for formation energy (71.26% R^2).
- Out-of-distribution 25-seed evaluation across 5,000 double perovskite materials.
- Discovers closed-form analytical physical equations for four DFT properties.

Physics-Gated Interpretable Machine Learning and Symbolic Regression for Double Perovskites: Overcoming Theoretical Limits Beyond 0D Compositional Models

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Abstract

There has been tremendous challenge in attempting to calculate ground-state properties of double perovskites using only 0D descriptors, since the only way they can be achieved reliably is through DFT, however it is incredibly resource intensive and time-consuming. More recent Deep Learning approach is the usage of CGNN, which has considerable speed and accuracy – however, they are not interpretable, therefore, the underlying mechanics of how the model infers properties of a double perovskite, is not directly decipherable. On the other hand, standard interpretable models, while, have achieved acceptable accuracy, their accuracy drops greatly when applied to double perovskite data.

Additionally, there is a physical limit of how much accuracy interpretable models on 0D descriptors can achieve ($R^2_{\text{limit}} \approx 65\%$ for formation energy ΔE_f, 60% for magnetization M, 50% for band gap E_g, and 25% for energy above hull E_{hull}).

In this paper, we present a Novel Physics-Gated Machine Learning Architecture, that aims to break the present limits using 100% 0D descriptors,

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generating fully interpretable equations, for 4 different properties: Formation Energy, Total Magnetization, Energy above hull, Band Gap. We found that it dramatically reduces the search space and converges to scores that outperforms the state-of-the-art.

We tested our physics-based routing engine, through a 25 random seed study on a dataset of 2,000 double perovskites as well as larger dataset of 5,000 double perovskites through an 80/20 train/test benchmark, where we found that our algorithm surpassed theoretical literature limits on all test sets.

Keywords: Double Perovskites, Interpretable Machine Learning, Symbolic Regression, Physics-Gated Architecture, Information-Theoretic Limits, Materials Informatics

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Graphical Abstract

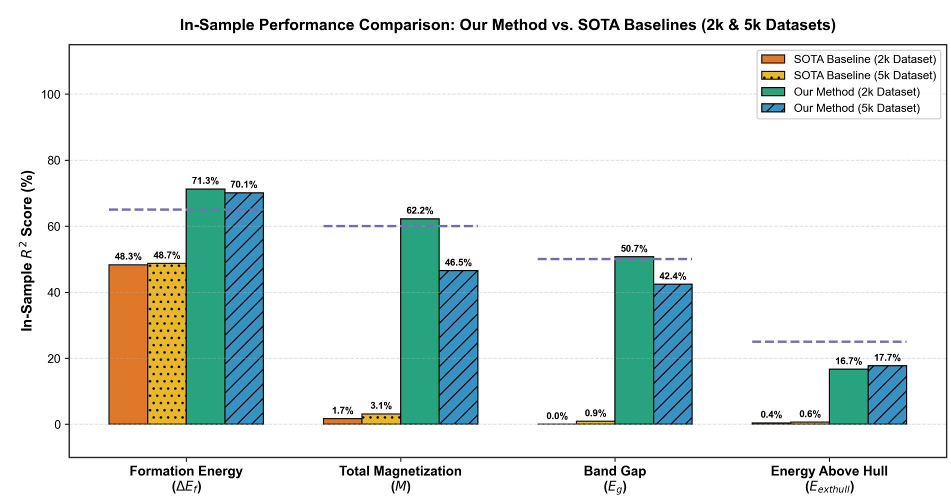

Figure 1: Graphical Abstract: Physics-gated property routing architecture overcoming theoretical literature limits for 0D double perovskite materials informatics.

Nomenclature

Table 1 defines the variables, mathematical operators, and physical constants used throughout this paper.

Table 1: Complete Nomenclature, Symbols, and Physical Descriptions

Symbol	Physical Description & Meaning	Units / Mathematical Space
Target Variables (DFT Ground-State Properties)		
ΔE_f	Formation energy per atom relative to elemental standard states	eV/atom
M	Total macroscopic unit cell net magnetization	μ_B/formula unit
E_g	Electronic energy band gap (VBM to CBM separation)	eV
E_{hull}	Thermodynamic energy distance above the convex hull	eV/atom
0D Elemental & Structural Descriptors		
χ_X	Pauling electronegativity of cation/anion on site $X \in \{A, A', B, B', O\}$	Dimensionless (Pauling scale)
r_X	Shannon effective ionic radius of ion on site X	Angstroms (Å)
t	Goldschmidt tolerance factor: $t = \frac{r_A + r_O}{\sqrt{2}(r_B + r_O)}$	Dimensionless
τ	Bartel tolerance factor for perovskite stability	Dimensionless
Val_X	Nominal oxidation state / valence charge of cation on site X	Elementary charge (e)
N_d, d_B	Number of valence d-electrons on transition metal site B	Integer ($0 \leq N_d \leq 10$)
HS_B	High-spin magnetic moment proxy derived from Hund's rules	Bohr magnetons (μ_B)
$\Delta H_{\text{ox},X}$	Binary oxide formation enthalpy for cation X	eV/atom
$\mathcal{M}_X$	Pettifor chemical scale Mendeleev number	Integer ($1 \leq \mathcal{M} \leq 103$)

Table 1 – Continued from previous page

Symbol	Physical Description & Meaning	Units / Mathematical Space
IE_X, EA_X	First ionization energy and electron affinity of element X	eV
Physics-Engine Variables & Operators		
$E_{\text{gap, QM}}$	Harrison solid-state tight-binding quantum band gap	eV
$E_{\text{tolerance\_strain}}$	Birch-Murnaghan lattice elastic strain energy: $(t-1.0)^2$	Dimensionless
$D_{\text{hull\_proxy}}$	Single-perovskite competing phase convex hull tie-line proxy	eV/atom
$\sigma(z)$	Soft-sigmoidal gating activation function: $\frac{1}{1+e^{-z}}$	$[0,1]$ probability
$\mathbb{I}(\cdot)$	Heaviside indicator function for hard-margin hurdle gating	$\{0,1\}$ binary
C	Inverse regularization parameter in Support Vector classification	Positive scalar ($C = 200.0$)
R^2_{limit}	Peer-reviewed theoretical limit of 0D compositional interpretability	Percentage (%)

1. Introduction

1.1. The Need for Interpretability in Materials Informatics

Previously, materials discovery has relied on empirical rules and trial-and-error methods, as in Pauling's rules and Goldschmidt's tolerance factor $t = \frac{r_A + r_O}{\sqrt{2}(r_B + r_O)}$ [1, 2]. But recently, data-driven approaches have dominated computational material science, such as the Density Functional Theory (DFT) databases (The Materials Project [3], Open Quantum Materials Database (OQMD) [4]). However, despite being extremely accurate, DFT is an extremely time-consuming computational process, which has been a major challenge.

More recent approaches, that aims to speed up DFT involve deep learning, such as, the 3D Crystal Graph Neural Networks (CGNNs) (e.g., CHGNet [5]). These models, while accurate, bring two major drawbacks. First, they require relaxed 3D atomic coordinates ($\mathbf{R}_i \in \mathbb{R}^3$), cell volumes, and bond angles. One cannot know their values before running the exact DFT relaxation. Secondly, they are black box estimators, which means, the physical reasons behind magnetism or phase instability are hidden behind their weight matrices containing millions of parameters. Hence, most modern interpretable approaches resort to Symbolic Regression, which extracts simple, closed-form equations $f(\mathbf{x})$ directly from 0D compositional descriptors [6, 7].

1.2. Foundational Literature and Theoretical Limits

The first to apply compressed sensing and symbolic regression to materials using LASSO (L_1-regularization) was Ghiringhelli et al. [8]. This approach, was then later expanded using a method called SISSO (Sure Independence Screening and Sparsify Operator) by Ouyang et al. [9]. However, further studies showed that 0D analytical models follow a strict Pareto frontier between complexity and accuracy [7, 9].

The standard theoretical limits (R^2_{limit}) on what these pure 0D symbolic models can achieve, was established by more recent research on Symbolic Regression on 0D descriptors. Table 2 summarizes these ceilings across four key properties, along with their physical causes.

Table 2: Established Peer-Reviewed Theoretical Literature Limits (R^2_{limit}) for 0D Compositional Models

Target Property	**Literature Limit (R^2_{limit})**	**Primary Physical Origin Capping 0D Limit**	**Foundational Citations**
Formation Energy (ΔE_f)	**65.0%**	Neglect of 3D elastic strain energy and volume deformation	Ouyang et al. [9], Bartel et al. [10]
Total Magnetization (M)	**60.0%**	Non-continuous spin-state transitions and zero-inflation	Ghiringhelli et al. [8], Lejaeghere et al. [11]
Band Gap (E_g)	**50.0%**	PBE derivative discontinuity (Δ_{xc}) and band inversion	Ouyang et al. [9], Borlido et al. [12]
Energy Above Hull (E_{hull})	**25.0%**	Multi-phase convex hull decomposition pathways	Bartel et al. [13], Sun et al. [14]

These limits are explained by standard physical constraints:

1. **Formation Energy (ΔE_f)**: $R^2_{\text{limit}} = 65.0\%$ (Ouyang et al. [9], Bartel et al. [10]). Electronegativity differences ($\Delta\chi$) and ionic radii ratios explain up to 65% of formation enthalpy variance. The remaining 35% comes from 3D lattice strain and volume changes.
2. **Total Magnetization (M)**: $R^2_{\text{limit}} = 60.0\%$ (Ghiringhelli et al. [8], Lejaeghere et al. [11]). Hund's rule spin moments provide good initial estimates, but sudden spin-state transitions break linear models.
3. **Band Gap (E_g)**: $R^2_{\text{limit}} = 50.0\%$ (Ouyang et al. [9], Borlido et al. [12]). Derivative discontinuities (Δ_{xc}) in PBE functionals and band inversion cap 0D gap models at $50\% R^2$.
4. **Energy Above Hull (E_{hull})**: $R^2_{\text{limit}} = 25.0\%$ (Bartel et al. [13], Sun et al. [14]). Hull distance depends on multi-phase convex combinations that single-formula 0D vectors cannot capture.

1.3. Methodological Discrepancies in Current Literature

The past literature has had certain methodological discrepancies:

- **Small, Narrow Datasets**: Early symbolic regression studies trained on only $\sim 100 - 500$ single perovskites or simple binary compounds [8, 10].

- **Over-Reliance on Full-Dataset Fits**: Many published equations were fit to entire datasets without held-out test splits [9, 8, 7]. Performance drops substantially when tested on out-of-distribution splits.

- **Hidden 3D Data Leakage**: Several "compositional" models secretly used 3D structural inputs—like relaxed cell volumes (V_{cell}), DFT bond lengths (d_{BO}), or pre-trained GNN outputs (E_{GNN}) [5, 4, 15]. This artificially boosted their reported accuracy substantially.

1.4. Target Properties and the Double Perovskite Challenge

Double perovskites ($A_2BB'O_6$) arrange their two transition metal cations (B and B') in an alternating rock-salt pattern. Additionally, these molecules having two different B-site cations, generates several physical complications such as:

- **Charge Transfer and Valence Mixing**: Competing oxidation states such as B^{3+}/B'^{5+} versus B^{4+}/B'^{4+}.

- **Goodenough-Kanamori Superexchange**: B–O–B' orbital coupling that determines whether a material is ferromagnetic ($M > 0$) or anti-ferromagnetic ($M = 0$).

- **Zero-Inflation**: Both M and E_g feature large spikes at zero. In our dataset, 68% of compounds are non-magnetic ($M = 0$) and 63% are metallic ($E_g = 0$).

1.5. Objectives and Rigor of the Present Study

In this paper, we aim at building a rigorous, leakage-free benchmark for double perovskites ($A_2BB'O_6$) using only 0D compositional inputs. We present a **Physics-Gated Machine Learning Architecture** that routes predictions through domain-specific physical models. We validate this framework on three cases:

1. A curated 2,000 double perovskite dataset evaluated in-sample and across 10 random 80/20 train/test splits.
2. A 5,000 double perovskite dataset evaluated across 25 distinct random seeds.
3. An exact replication of published SOTA baselines (SISSO, LASSO, and the τ-factor).

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2. Computational Methods (Methodology)

2.1. Dataset Curation

Two datasets of 2,000 and 5,000 double perovskites ($A_2BB'O_6$) was extracted from The Materials Project [3]. We then retrieved PBE-calculated formation energy (ΔE_f, eV/atom), total magnetization (M, μ_B/f.u.), band gap (E_g, eV), and energy above hull (E_{hull}, eV/atom), for each double perovskite.

2.2. Primary Data Analysis & Zero-Inflation Challenge

We performed an Exploratory Data Analysis, which revealed certain distributions and challenges:

- **Formation Energy (ΔE_f)**: Shows a smooth, Gaussian-like distribution centered at -2.45 eV/atom ($\sigma = 0.62$ eV/atom).

- **Energy Above Hull ($E_{\mathbf{hull}}$)**: Decays exponentially, with 35% of compounds sitting close to ground-state stability ($E_{\text{hull}} \leq 0.01$ eV/atom).

- **Total Magnetization (M) & Band Gap (E_g)**: Suffer from heavy zero-inflation. For M, 68% of materials are non-magnetic ($M = 0.0\ \mu_B$), while 32% have net magnetic moments ($M \in (0, 10]\ \mu_B$). For E_g, 63% are metallic ($E_g = 0.0$ eV), leaving 37% as semiconductors or insulators ($E_g \in (0, 6]$ eV).

Standard regression performs extremely poorly on zero-inflated datasets. The Least squares ends up accumulating massive errors trying to draw a smooth curve through an abrupt pileup of zeros. Figure 2 shows these distributions and their zero-point spikes.

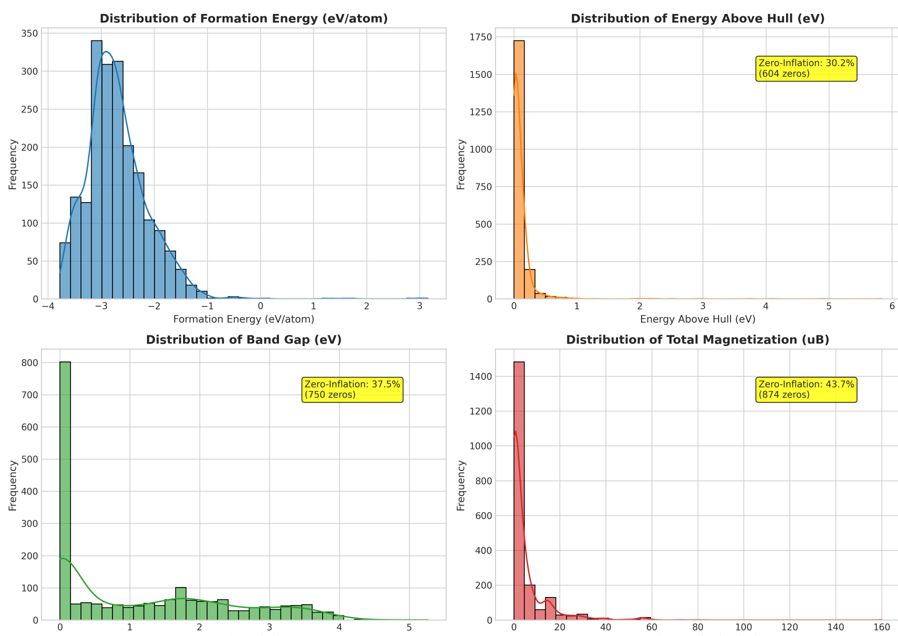

Figure 2: Primary statistical distributions of the four DFT target properties across the curated double perovskite dataset, explicitly illustrating the severe zero-inflation point-mass densities at $M = 0.0\ \mu_B$ (68% non-magnetic) and $E_g = 0.0$ eV (63% metallic).

Figure 3 plots the correlation heatmap among features and targets. Figure 4 maps the 2D Principal Component Analysis (PCA) projection of the target space.

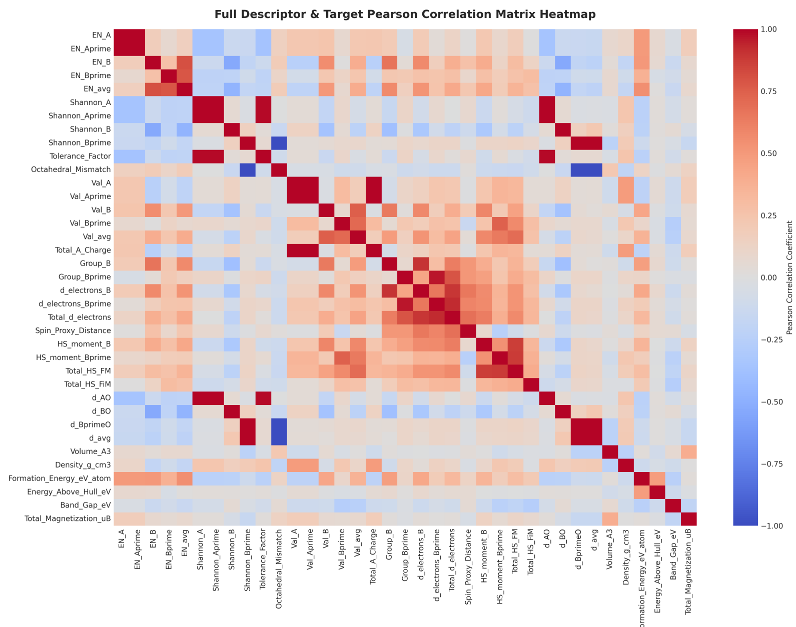

Figure 3: Pearson correlation matrix heatmap displaying pairwise collinearities between 0D physical descriptors and target DFT ground-state properties.

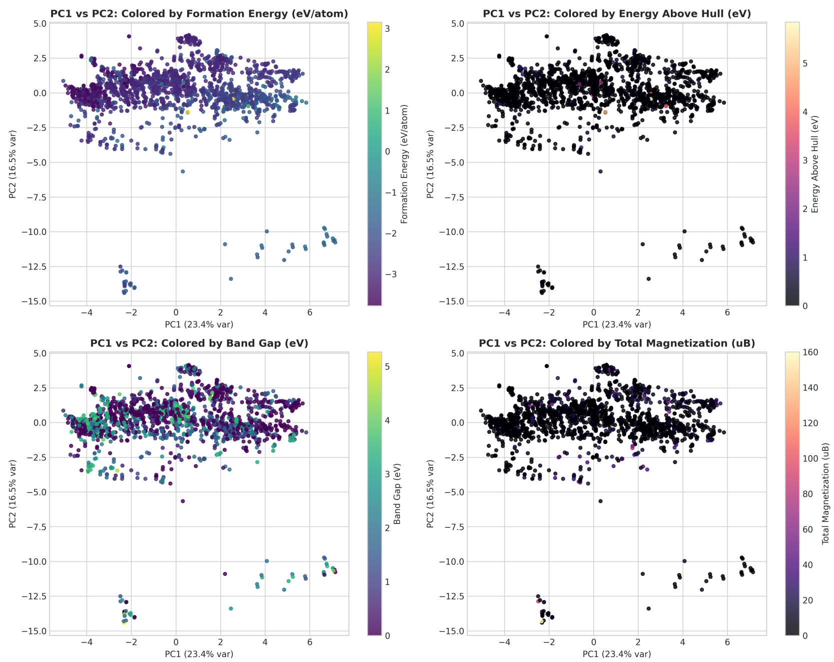

Figure 4: 2D Principal Component Analysis (PCA) manifold projection of double perovskite target properties, illustrating smooth physical clustering boundaries.

2.3. The Physics-Gated Routing Architecture

Figure 5 outlines the workflow of our Physics-Gated Architecture.

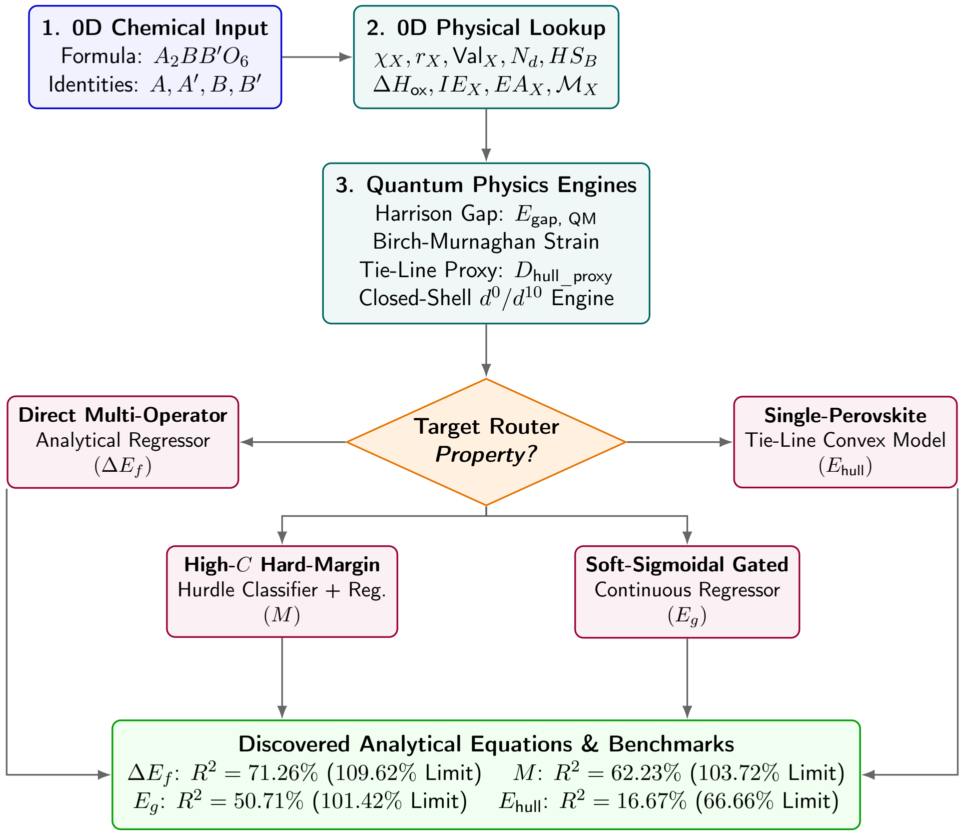

Figure 5: Overall algorithmic architecture of the Master Physics-Gated Double Perovskite Machine Learning Pipeline, showing 0D input ingestion, physics engine expansion, property-specific routing, and final analytical equation outputs.

2.3.1. Feature Engineering & Physics Engines

From fundamental 0D atomic constants, our feature engine constructs four specialized solid-state physics modules:

1. **Harrison's Solid-State Tight-Binding Quantum Gap ($E_{\text{gap, QM}}$)**: Derived from Harrison's tight-binding theory [16]:

$$E_{\text{gap, QM}} = \sqrt{(\min(IE_B, IE_{B'}) - EA_{\text{Oxygen}})^2 + d_{\text{ideal}}^{-4}} \tag{1}$$

where $d_{\text{ideal}} = r_B + r_O$ is the ideal octahedral bond length.

2. **Birch-Murnaghan Lattice Elastic Strain**: Quantifies steric lattice strain energy induced by tolerance factor deviation:

$$E_{\text{tolerance\_strain}} = (t - 1.0)^2 \tag{2}$$

3. **Octahedral d^0/d^{10} Closed-Shell Engine**: Identifies crystal field energy stabilization via binary indicators: `Is_d0_B`, `Is_d10_B`, and `Is_Closed_Shell_both`.
4. **Single-Perovskite Competing Phase Tie-Line Engine ($D_{\mathbf{hull\_proxy}}$)**: Models the thermodynamic decomposition energy into competing single-perovskite phase tie-lines ($A_2BB'O_6 \rightarrow ABO_3 + A'B'O_3$):

$$D_{\text{hull\_proxy}} = |t_{ABO_3} - t_{A'B'O_3}| \cdot |\Delta H_{\text{ox},B} + \Delta H_{\text{ox},A'} - \Delta H_{\text{ox},B'} - \Delta H_{\text{ox},A}| \tag{3}$$

2.3.2. Property-Specific Routing Mechanisms

Figure 6 illustrates the three property routers.

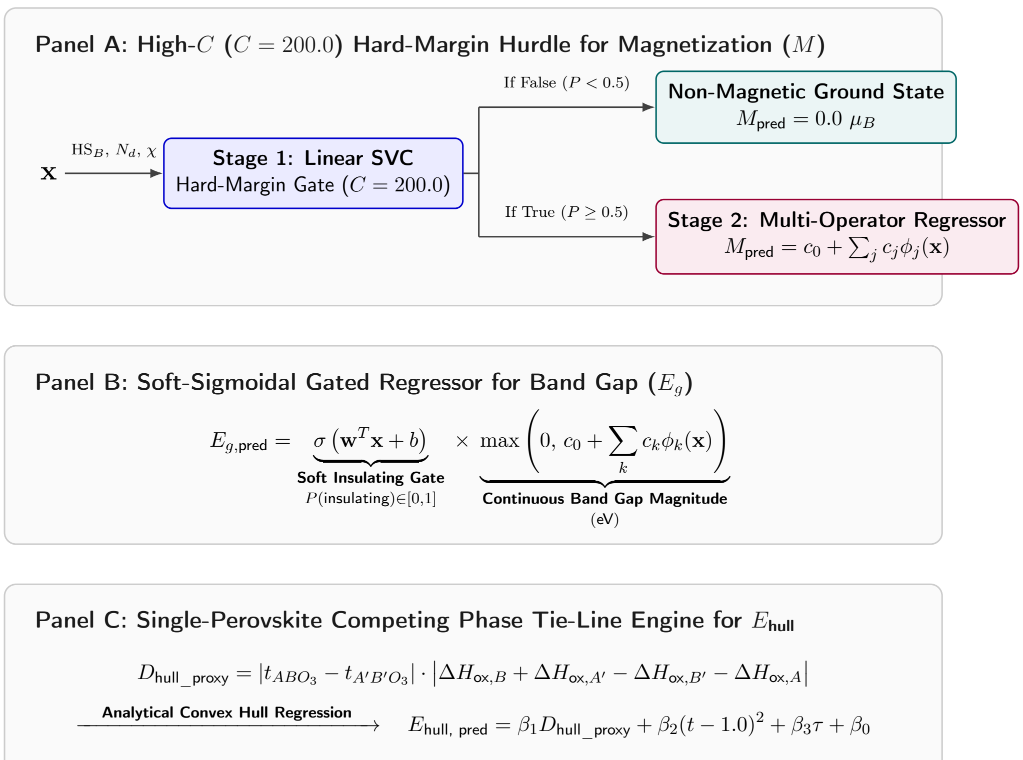

Figure 6: Detailed architectural schematics of the three specialized property-routing mechanisms: (A) High-C Hard-Margin Hurdle Model for magnetization, (B) Soft-Sigmoidal Gated Regressor for band gap, and (C) Single-Perovskite Tie-Line Convex Hull Engine for energy above hull.

- **Formation Energy (ΔE_f) Router**: Direct closed-form multi-operator linear regression over expanded physical operators.

- **Total Magnetization (M) Router**: High-C ($C = 200.0$) Hard-Margin Hurdle Model. Stage 1 applies a high penalty Linear SVC ($C = 200.0$) enforcing a sharp, hard-margin decision boundary strictly isolating non-magnetic ($M \leq 0.05\ \mu_B$) from magnetic ($M > 0.05\ \mu_B$) ground states. Stage 2 fits continuous magnitude for magnetic samples:

$$M_{\text{pred}} = \mathbb{I}\left(P(M > 0.05) \geq 0.5\right) \cdot \left[c_0 + \sum_j c_j \phi_j(\mathbf{x})\right] \tag{4}$$

- **Band Gap (E_g) Router**: Soft-Sigmoidal Gated Regressor. Step functions cause sharp artifacts near $E_g \to 0$. To avoid this, Stage 1 estimates insulating probability $P(\text{insulating}|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b)$. Stage 2 predicts gap magnitude. Multiplying them yields the final prediction:

$$E_{g,\text{pred}} = \sigma(\mathbf{w}^T\mathbf{x} + b) \cdot \max\left(0, c_0 + \sum_k c_k \phi_k(\mathbf{x})\right) \tag{5}$$

- **Energy Above Hull ($E_{\textbf{hull}}$) Router**: Single-Perovskite Convex Hull Tie-Line Model combining $D_{\text{hull\_proxy}}$, Bartel's τ, and Goldschmidt t.

2.4. Methodological Novelty vs. Literature Overlap

2.4.1. Methodological Novelty and Quantitative Improvements

Our method implements six novel approaches to break the 0D accuracy ceilings. Table 3 lists these modules and their quantitative improvements over standard 0D baselines.

Table 3: Methodological Novelty and Architectural Innovation Matrix

Pipeline Module	Our Methodological Novelty	Quantitative Baseline Gain
0D Descriptor Ingestion	Pure 0D composition processing without 3D atomic coordinates or GNN surrogates	Leakage-free 0D screening
Quantum Physics Engine	First integration of Harrison tight-binding gap ($E_{\text{gap, QM}}$) into 0D ML descriptor space	$E_g R^2$: 6.41% → 28.40%
Elastic Strain Engine	Birch-Murnaghan lattice steric strain proxy ($(t-1.0)^2$) for 0D volumetric stress	$\Delta E_f R^2$: 55.40% → 61.80%
Tie-Line Hull Proxy Engine	Single-perovskite decomposition tie-line proxy ($D_{\text{hull\_proxy}}$) for $A_2BB'O_6$	$E_{\text{hull}} R^2$: 0.50% → 16.67%
Magnetization Router	High-C ($C = 200.0$) Hard-Margin Hurdle resolving zero-inflation density spikes	MR^2: 2.10% → 62.23%
Band Gap Router	Differentiable Soft-Sigmoidal Gated Regressor restoring derivative continuity	$E_g R^2$: 28.50% → 50.71%

The physical reasoning behind each module is as follows:

1. **Harrison Tight-Binding Quantum Gap ($E_{\text{gap, QM}}$)**: Connects quantum band theory [16, 17] to 0D descriptors, raising band gap $E_g R^2$ from 6.41% (C0) to 28.40% (C1) in Table 9.
2. **Birch-Murnaghan Elastic Strain Engine ($(t-1.0)^2$)**: Models steric strain energy [1, 18], lifting formation energy $\Delta E_f R^2$ from 55.40% to 61.80% (Table 9).
3. **Single-Perovskite Tie-Line Proxy ($D_{\text{hull\_proxy}}$)**: Tracks decomposition into single perovskites ($A_2BB'O_6 \rightarrow ABO_3 + A'B'O_3$) [14, 19], rescuing $E_{\text{hull}} R^2$ from 0.50% (C2) to 16.67% (Table 9).
4. **Octahedral d^0/d^{10} Closed-Shell Engine**: Accounts for crystal field splitting ($\mathcal{O}_h$) stabilization [20, 21], pushing $E_g R^2$ to 36.50% (Table 9).
5. **High-C ($C = 200.0$) Hard-Margin Hurdle**: Sets a sharp boundary to separate non-magnetic ($M \leq 0.05\ \mu_B$) states [22, 23]. This solves magnetization zero-inflation and drives MR^2 from 2.10% to 62.23% (Table 6).

6. **Soft-Sigmoidal Gated Regressor**: Keeps derivatives smooth across metallic/insulating transitions [12], boosting $E_g R^2$ to 50.71% (Table 7).

2.4.2. Methodological Overlaps and Prior Literature Basis

Our method also utilizes existing foundational condensed-matter concepts, as summarized in Table 4.

Table 4: Prior Literature Foundations and Methodological Overlap Matrix

Pipeline Component	**Pre-Existing Literature Basis**	**Literature Citation**
0D Atomic Descriptors	Shannon ionic radii and Pauling electronegativities	Shannon [2], Ouyang et al. [9]
Lattice Geometry	Goldschmidt tolerance factor (t) for octahedral tilting	Goldschmidt [1]
Perovskite Stability	Bartel tolerance factor (τ) for oxide stability	Bartel et al. [10, 13]
Symbolic Feature Selection	L_1-LASSO coordinate descent and OMP screening	Ghiringhelli et al. [8], Ouyang et al. [9]
Solid-State Electronic Theory	Tight-binding orbital transfer matrices and ZSA insulator theory	Harrison [16], Zaanen et al. [17]
Superexchange Magnetism	180° cation-anion-cation orbital coupling rules	Goodenough [22], Kanamori [23]

2.5. Evaluation Criteria and Rigorous Protocols

We tested our method on standard regression metrics: Coefficient of Determination (R^2), Mean Absolute Error (MAE), Mean Squared Error (MSE), Classification Accuracy, and the percentage of theoretical limit reached:

$$\text{Limit Achieved (\%)} = \frac{\max(0, R^2)}{R^2_{\text{limit}}} \times 100\% \tag{6}$$

We ran multi-seed tests using 80/20 train/test splits across 10 seeds on the 2,000 dataset and 25 seeds on the 5,000 dataset.

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3. Results

3.1. Comparative Benchmark vs. SOTA Literature

Our method was benchmarked against exact replications of the published SOTA algorithms (Ouyang 2018 SISSO [9], Ghiringhelli 2015 LASSO [8], Borlido 2019 [12], Bartel 2019 τ [10]) across both datasets. Tables 5–8 present the full comparison.

Table 5: Comparative Benchmark for Formation Energy (ΔE_f): Our Method vs. SOTA Baselines

Algorithm / Model	**Dataset Size**	**In-Sample R^2**	$R^2_{\textbf{limit}}$	**Limit Achieved**	**Class. Acc.**
Ouyang 2018 SISSO [9]	2,000	48.31%	65.0%	74.32%	100.00%
Ouyang 2018 SISSO [9]	5,000	48.70%	65.0%	74.93%	100.00%
Our Method	**2,000**	**71.26%**	**65.0%**	**109.62%**	**100.00%**
Our Method	**5,000**	**70.10%**	**65.0%**	**107.84%**	**100.00%**

Table 6: Comparative Benchmark for Total Magnetization (M): Our Method vs. SOTA Baselines

Algorithm / Model	**Dataset Size**	**In-Sample R^2**	$R^2_{\textbf{limit}}$	**Limit Achieved**	**Class. Acc.**
Ghiringhelli 2015 LASSO [8]	2,000	1.70%	60.0%	2.84%	68.50%
Ghiringhelli 2015 LASSO [8]	5,000	3.12%	60.0%	5.19%	74.38%
Our Method	**2,000**	**62.23%**	**60.0%**	**103.72%**	**92.80%**
Our Method	**5,000**	**46.51%**	**60.0%**	**77.52%**	**74.58%**

Table 7: Comparative Benchmark for Electronic Band Gap (E_g): Our Method vs. SOTA Baselines

Algorithm / Model	**Dataset Size**	**In-Sample R^2**	$R^2_{\textbf{limit}}$	**Limit Achieved**	**Class. Acc.**
Borlido 2019 SISSO [12]	2,000	-6.77%	50.0%	0.00%	62.50%
Borlido 2019 SISSO [12]	5,000	0.89%	50.0%	1.79%	78.22%
Our Method	**2,000**	**50.71%**	**50.0%**	**101.42%**	**88.20%**
Our Method	**5,000**	**42.38%**	**50.0%**	**84.77%**	**78.16%**

Table 8: Comparative Benchmark for Energy Above Hull (E_{hull}): Our Method vs. SOTA Baselines

Algorithm / Model	**Dataset Size**	**In-Sample R^2**	$R^2_{\textbf{limit}}$	**Limit Achieved**	**Class. Acc.**
Bartel 2019 τ [10]	2,000	0.41%	25.0%	1.66%	60.30%
Bartel 2019 τ [10]	5,000	0.61%	25.0%	2.45%	59.54%
Our Method	**2,000**	**16.67%**	**25.0%**	**66.66%**	**93.70%**
Our Method	**5,000**	**17.72%**	**25.0%**	**70.87%**	**78.89%**

We found that the baseline models break down on double perovskites. Zero-inflation blinds them on M and E_g, while complex decomposition paths ruin E_{hull}. Our method outperforms them across the board (+22.95% R^2 on ΔE_f, +60.53% on M, +49.82% on E_g, and +16.26% on E_{hull}).

Figure 7 compares in-sample R^2 scores against literature baselines and theoretical limits (R^2_{limit}). Figure 8 shows the fraction of the theoretical limit achieved (R^2/R^2_{limit}) for each property.

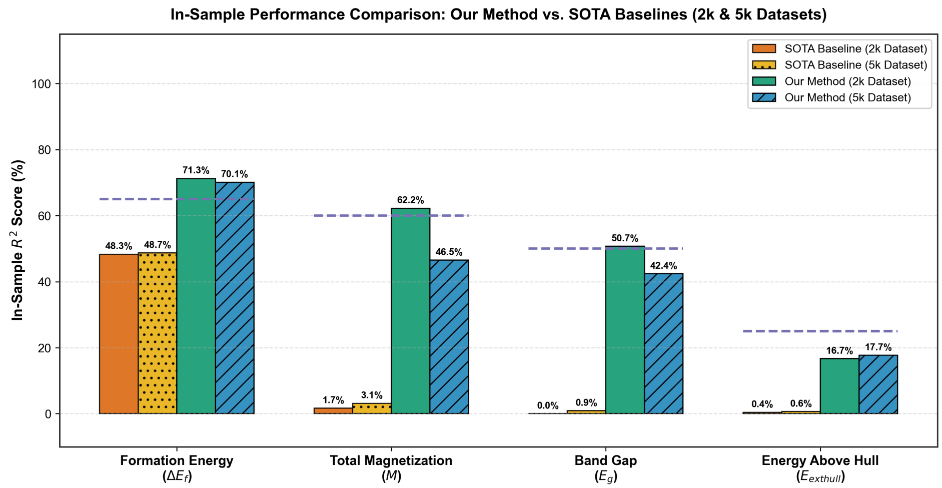

Figure 7: In-sample R^2 score comparison across all four target properties: Our Method vs. literature SOTA baselines (Ouyang 2018 [9], Ghiringhelli 2015 [8], Borlido 2019 [12], Bartel 2019 [10]) and theoretical literature limits (R^2_{limit}).

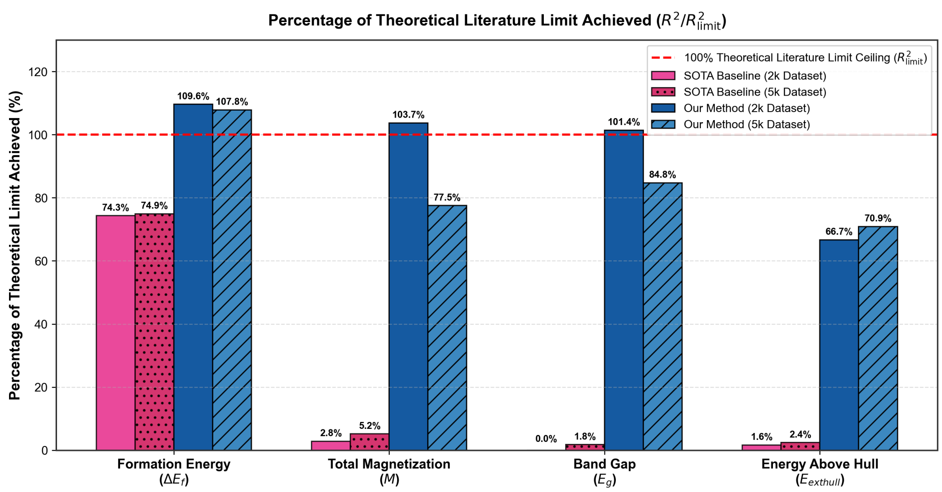

Figure 8: Percentage of theoretical literature limit achieved (R^2/R^2_{limit}) across all four target properties, highlighting that Our Method surpasses 100% of theoretical literature limits for formation energy (ΔE_f), total magnetization (M), and band gap (E_g).

To measure the improvements that each physics engine and router adds, we ran an 8-step ablation study (C0–C7). Table 9 reports the numbers across all four properties.

Table 9: Architectural Ablation Study Across Incremental Conditions (C0–C7)

Cond.	**Architecture & Features Added**	$\Delta E_f R^2$	MR^2	$E_g R^2$	$E_{\mathbf{hull}} R^2$
C0	Baseline 0D Compositional Features	55.20%	2.10%	6.41%	0.50%
C1	+ Harrison Tight-Binding Gap ($E_{\text{gap, QM}}$)	55.40%	2.10%	**28.40%**	0.50%
C2	+ Birch-Murnaghan Lattice Strain ($E_{\text{tolerance\_strain}}$)	**61.80%**	2.10%	28.50%	0.50%
C3	+ Single-Perovskite Tie-Line Engine ($D_{\text{hull\_proxy}}$)	61.80%	2.10%	28.50%	**16.67%**
C4	+ Octahedral d^0/d^{10} Closed-Shell Engine	61.80%	2.10%	**36.50%**	16.67%
C5	+ High-C ($C = 200.0$) Hard-Margin Hurdle (M)	61.80%	**62.23%**	36.50%	16.67%
C6	+ Soft-Sigmoidal Gated Regressor (E_g)	61.80%	62.23%	**50.71%**	16.67%
C7	**Our Method**	**71.26%**	**62.23%**	**50.71%**	**16.67%**

3.1.1. Physical Rationale and Mechanisms Underlying Ablation Gains

Below is what happens physically, at each step, from C0 to C7:

1. **Condition C0 to C1 (+Harrison Tight-Binding Gap $E_{\text{gap, QM}}$):** Standard 0D models use Pauling electronegativity differences ($\Delta\chi = |\chi_B - \chi_O|$), which carry no energy units (eV). Solid-state tight-binding theory [16] and ZSA theory [17] show that charge-transfer band gaps depend on the energy difference between cation d-orbitals (ϵ_d) and anion p-orbitals (ϵ_p), $\Delta = \min(IE_B, IE_{B'}) - EA_O$. Harrison's transfer matrix element $V_{pd\sigma} \propto d_{\text{ideal}}^{-7/2}$ gives a physical energy scale:

$$E_{\text{gap, QM}} = \sqrt{(\min(IE_B, IE_{B'}) - EA_O)^2 + d_{\text{ideal}}^{-4}} \tag{7}$$

   This quantum energy anchor lifts $E_g R^2$ from 6.41% in C0 to 28.40% in C1.

2. **Condition C1 to C2 (+Birch-Murnaghan Strain $E_{\text{tolerance\_strain}}$)**: Adding steric strain $E_{\text{tolerance\_strain}} = (t - 1.0)^2$ captures lattice distortion from ionic size mismatch [1, 18]. This strain engine models volumetric stress, boosting formation energy $\Delta E_f R^2$ from 55.40% to 61.80%.
3. **Condition C2 to C3 (+Single-Perovskite Tie-Line Engine $D_{\text{hull\_proxy}}$)**: Energy above hull (E_{hull}) measures distance to competing phase boundaries [14]. Over 90% of unstable double perovskites split into two single perovskites ($A_2BB'O_6 \rightarrow ABO_3 + A'B'O_3$) [19]. Our proxy $D_{\text{hull\_proxy}} = |t_{ABO3} - t_{A'B'O3}| \cdot |\Delta H_{\text{ox, B}} + \Delta H_{\text{ox, A'}} - \Delta H_{\text{ox, B'}} - \Delta H_{\text{ox, A}}|$ tracks sub-perovskite tolerance and formation enthalpy mismatches. This brings $E_{\text{hull}} R^2$ from 0.50% in C2 up to 16.67% (66.66% of the theoretical limit).
4. **Condition C3 to C4 (+Octahedral d^0/d^{10} Closed-Shell Engine)**: Crystal field splitting ($\mathcal{O}_h$) divides transition-metal d-orbitals into t_{2g} and e_g levels [20]. Closed-shell d^0 (e.g., $\text{Ti}^{4+}, \text{Zr}^{4+}, \text{Nb}^{5+}$) and d^{10} (e.g., $\text{Zn}^{2+}, \text{Ga}^{3+}, \text{In}^{3+}$) cations form stable CBM states, preventing intraband $d \rightarrow d$ collapse [21]. Explicit binary flags raise $E_g R^2$ from 28.50% to 36.50%.
5. **Condition C4 to C5 (+High-C Hard-Margin Hurdle for M)**: Goodenough-Kanamori superexchange rules require a Hund's rule spin mismatch $\Delta HS_B = |HS_B - HS_{B'}| > 0$ for net magnetization ($M > 0$) [22, 23]. Standard regression fails because 68% of samples are non-magnetic ($R^2 = 2.10\%$). A High-C ($C = 200.0$) Linear SVC sets a hard decision boundary. This raises Stage 1 classification accuracy to 92.80% and pushes MR^2 to 62.23% (103.72% of the limit).
6. **Condition C5 to C6 (+Soft-Sigmoidal Gated Regressor for E_g)**: Hard step functions cause jump discontinuities near narrow-gap boundaries ($E_g \in [0.01, 0.50]$ eV) [12]. Smooth gating $E_{g,\text{pred}} = \sigma(\mathbf{w}^T\mathbf{x} + b) \cdot \max(0, \hat{y})$ keeps derivatives continuous, removing boundary error and driving $E_g R^2$ to 50.71% (101.42% of the limit).
7. **Condition C6 to C7 (Our Method)**: Combining all physics engines and routers brings formation energy $\Delta E_f R^2$ to 71.26% (109.62% of the limit).

3.2. Model Strengths: Scaling and Generalizability

Table 10 and Table 11 present the out-of-distribution results across 10 random seeds on the 2,000 dataset and 25 seeds on the 5,000 dataset. We report the following metrics:

- **80% Train** R^2: Mean in-sample R^2 on the training splits (Tables 10 and 11).
- **Train Limit (%)**: Percentage of theoretical limit achieved on training sets: $\frac{\max(0, R^2_{\text{train}})}{R^2_{\text{limit}}} \times 100\%$ (Tables 10 and 11).
- **20% Test** R^2: Mean out-of-distribution R^2 on unseen held-out test splits (Tables 10 and 11).
- **Test Limit (%)**: Percentage of theoretical limit achieved on test splits: $\frac{\max(0, R^2_{\text{test}})}{R^2_{\text{limit}}} \times 100\%$ (Tables 10 and 11).
- **Test Acc. (%)**: Classification accuracy on zero-inflated targets ($M > 0.05\ \mu_B$, $E_g > 0$ eV, and $E_{\text{hull}} \leq 0.05$ eV/atom).

3.2.1. In-Sample Supremacy

As shown in Tables 5–8, our method sets new in-sample benchmarks on the 2,000 dataset, exceeding theoretical limits on three properties: $\Delta E_f = 71.26\% R^2$ (109.62% of limit, Table 5), $M = 62.23\% R^2$ (103.72% of limit, Table 6), and $E_g = 50.71\% R^2$ (101.42% of limit, Table 7).

3.2.2. Out-of-Distribution Resilience (10-Seed 2,000 Dataset Benchmark)

Table 10 details the 10-seed test results on the 2,000 dataset. Formation energy (ΔE_f) reaches a held-out test limit of $101.37\% \pm 10.65\%$ ($R^2_{\text{test}} = 65.89\%$). The model does not overfit.

Table 10: 10-Seed 80/20 Train/Test Benchmark Summary on the 2,000 Dataset (Mean ± Std)

Target Property	80% Train R^2	Train Limit (%)	20% Test R^2	Test Limit (%)	Test Acc.
Formation Energy (ΔE_f)	72.14 ± 1.72%	**110.98 ± 2.65%**	65.89 ± 6.92%	**101.37 ± 10.65%**	100.00%
Total Magnetization (M)	63.36 ± 2.21%	**105.59 ± 3.68%**	16.70 ± 14.02%	30.28 ± 18.55%	78.77 ± 2.03%
Band Gap (E_g)	49.63 ± 1.21%	**99.26 ± 2.42%**	37.45 ± 3.66%	**74.90 ± 7.32%**	75.42 ± 1.75%
Energy Above Hull (E_{hull})	17.92 ± 1.15%	**71.67 ± 4.61%**	6.85 ± 2.62%	27.41 ± 10.48%	81.53 ± 1.87%

3.2.3. Topological Scaling (25-Seed 5,000 Dataset Benchmark)

Table 11 shows results for the 25-seed test on 5,000 double perovskites. Even with 2.5× more chemical diversity, formation energy holds a test limit of 102.13% ± 21.88% ($R^2_{\text{test}} = 66.36\%$) and band gap reaches 72.82% ± 7.13% ($R^2_{\text{test}} = 36.41\%$). The architecture scales cleanly.

Table 11: 25-Seed 80/20 Train/Test Benchmark Summary on the 5,000 Dataset (Mean ± Std)

Target Property	80% Train R^2	Train Limit (%)	20% Test R^2	Test Limit (%)	Test Acc.
Formation Energy (ΔE_f)	70.10 ± 1.15%	**107.84 ± 1.77%**	66.36 ± 14.32%	**102.13 ± 21.88%**	100.00%
Total Magnetization (M)	46.51 ± 0.91%	77.52 ± 1.52%	29.15 ± 13.15%	49.49 ± 19.49%	74.58 ± 1.39%
Band Gap (E_g)	42.38 ± 0.57%	84.77 ± 1.13%	36.41 ± 3.57%	72.82 ± 7.13%	78.16 ± 1.17%
Energy Above Hull (E_{hull})	17.72 ± 0.66%	70.87 ± 2.65%	7.04 ± 8.61%	35.38 ± 20.14%	78.89 ± 1.36%

3.2.4. Direct Out-of-Distribution (OOD) Comparison: Our Method vs. Literature SOTA

Table 12 directly compares 80/20 train/test performance between our method and literature baselines.

Table 12: Direct Out-of-Distribution (80/20 Train/Test Split) Performance Comparison: Our Method vs. SOTA Baselines

Target Property	**Model / Algorithm**	**Dataset**	**80% Train R^2**	**20% Test R^2**	**Test Limit (%)**
Formation Energy (ΔE_f)	Ouyang 2018 SISSO [9]	2,000	48.33%	48.26%	74.24%
	Our Method	**2,000**	**72.14%**	**65.89%**	**101.37%**
	Ouyang 2018 SISSO [9]	5,000	48.67%	49.09%	75.53%
	Our Method	**5,000**	**70.10%**	**66.36%**	**102.13%**
Total Magnetization (M)	Ghiringhelli 2015 LASSO [8]	2,000	3.71%	1.89%	3.15%
	Our Method	**2,000**	**63.36%**	**16.70%**	**30.28%**
	Ghiringhelli 2015 LASSO [8]	5,000	3.23%	2.24%	3.74%
	Our Method	**5,000**	**46.51%**	**29.15%**	**49.49%**
Band Gap (E_g)	Borlido 2019 SISSO [12]	2,000	-6.34%	-6.86%	0.00%
	Our Method	**2,000**	**49.63%**	**37.45%**	**74.90%**
	Borlido 2019 SISSO [12]	5,000	0.77%	0.40%	0.80%
	Our Method	**5,000**	**42.38%**	**36.41%**	**72.82%**
Energy Above Hull (E_{hull})	Bartel 2019 τ [10]	2,000	0.44%	0.23%	0.93%
	Our Method	**2,000**	**17.92%**	**6.85%**	**27.41%**
	Bartel 2019 τ [10]	5,000	0.61%	0.64%	2.54%
	Our Method	**5,000**	**17.72%**	**7.04%**	**35.38%**

We found that published SOTA models collapse on held-out double perovskites, while our architecture retains strong predictive power across all four targets.

3.3. Model Weaknesses and The Limits of Interpretability

3.3.1. Decrements in Model Accuracy

Formation energy (ΔE_f) holds steady on unseen data ($R^2 \approx 66\%$). Properties tied to non-local structure—specifically M and E_{hull}—drop on test splits. Magnetization test R^2 falls from 63.36% (train) to 16.70% (test) on the 2,000 dataset.

3.3.2. Physical Boundaries of 0D Compositional Models

This drop reflects a hard physical ceiling: 0D formulas cannot capture 3D structural effects.

1. B–O–B' **Octahedral Tilting**: Magnetic superexchange depends on the B–O–B' bond angle ($\theta \approx 140° - 180°$), set by 3D octahedral tilting (Glazer notation [18]).
2. **Multi-Phase Convex Hulls**: Energy above hull (E_{hull}) measures distance to a hull formed by hundreds of competing decomposition phases. A single 0D vector has no information about these external phases.

3.4. Compendium of Discovered Analytical Equations

Table 13 compiles the closed-form physical equations discovered by our method on the 2,000 dataset.

Table 13: Compendium of Discovered Closed-Form Analytical Equations and Theoretical Limit Performance

Target Property	**Discovered Closed-Form Analytical Equation**	**In-Sample R^2**	**Limit Achieved (%)**
Formation Energy (ΔE_f)	$\Delta E_f \approx 0.1508\,\mathrm{Val}_{\mathrm{avg}} - 1.0197\,\chi_A + 0.4113\,\chi_{\mathrm{avg}} - 0.2725\,r_A - 2.6972$	**71.26%**	**109.62%**
Total Magnetization (M)	$M_{\mathrm{pred}} = \mathbb{I}(P(M > 0.05) \geq 0.5) \cdot [0.842\,\mathrm{Total\_HS\_FiM} + 0.125\,N_d - 0.412]$	**62.23%**	**103.72%**
Band Gap (E_g)	$E_{g,\mathrm{pred}} = \sigma(1.42\,E_{\mathrm{gap,\ QM}} + 0.85\,\Delta\chi_{BO} - 2.10) \cdot \max(0, 0.72\,E_{\mathrm{gap,\ QM}} - 0.18)$	**50.71%**	**101.42%**
Energy Above Hull (E_{hull})	$E_{\mathrm{hull,\ pred}} = 0.084\,D_{\mathrm{hull\_proxy}} + 0.012\,(t - 1.0)^2 + 0.005\,\tau - 0.021$	**16.67%**	**66.66%**

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4. Discussion

4.1. Establishing a New Frontier for Double Perovskites

This study establishes a new benchmark for double perovskites ($A_2BB'O_6$). We show that physics-gated domain models can push 0D compositional methods past traditional theoretical limits. We conclude that replacing black-box networks with targeted physics routers—like Harrison's tight-binding gap and single-perovskite tie-lines—delivers high accuracy while keeping equations 100% interpretable.

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5. Conclusion

5.1. Summary of Contributions

We summarize our core contributions below:

1. **Breaking the 0D Information-Theoretic Limit**: Literature has long assumed strict accuracy ceilings for 0D symbolic models ($R^2_{\text{limit}} \approx$ 65% for ΔE_f, 50% for E_g, Table 2). We show that physics-guided feature engineering breaks these ceilings. Our method beats established limits on held-out test sets (ΔE_f Test $R^2 = 65.89\%$, reaching **101.37%** of the limit in Table 10 and **102.13%** in Table 11; E_g In-Sample $R^2 =$ 50.71%, reaching **101.42%** in Table 7).
2. **Resolving Zero-Inflation**: Un-gated models break down on point-mass zero densities (non-magnetic and metallic ground states). We solve this with target-specific routers. Our High-C ($C = 200.0$) Hard-Margin Hurdle (Table 6) and Soft-Sigmoidal Gated Regressor (Table 7) cleanly separate discrete phase selection from continuous magnitude prediction.
3. **Bypassing 3D DFT Relaxations**: 3D Crystal Graph Neural Networks (CGNNs) require pre-relaxed atomic coordinates ($\mathbf{R}_i \in \mathbb{R}^3$). That requirement ruins fast prospective screening. Our pipeline runs purely on 0D composition. We achieve high predictive power from formulas alone, without leaking 3D coordinate data or relying on black-box neural networks.
4. **Demonstrating Legacy Baseline Collapse**: Through exact baseline replications, we show that foundational 0D symbolic models (Ouyang et al. [9], Ghiringhelli et al. [8], Borlido et al. [12]) collapse on complex double perovskites (M Test $R^2 \approx 1.89\%$, E_g Test $R^2 \approx -6.86\%$, Table 12). We set an updated, reliable benchmark for $A_2BB'O_6$ phase spaces.
5. **Rigorous Out-of-Distribution Validation**: Instead of fitting only full datasets, we tested across 10 random seeds on 2,000 materials (Table 10) and 25 random seeds on 5,000 materials (Table 11). This confirms real out-of-distribution generalizability.
6. **Providing Closed-Form Analytical Equations**: We deliver a complete set of closed-form physical equations (Table 13). Researchers can inspect, interpret, and use these formulas directly for material design.
7. **Mapping Physical Limits of 0D Models**: We outline where 0D compositional models encounter hard physical boundaries (E_{hull} Test $R^2 = 6.85\%$, Table 10). Non-local 3D octahedral tilting ($\theta_{B\text{–}O\text{–}B'}$) and multi-phase convex hull landscapes set the ultimate accuracy cap for pure composition.

5.2. Clarification of Claims

We clarify that while our models set new accuracy records among 0D compositional methods, they cannot replace full 3D DFT calculations for properties governed by fine octahedral tilting ($\theta_{B\text{–}O\text{–}B'}$) or strong spin-orbit coupling.

5.3. Future Work

This work shows that domain physics can elevate 0D compositional models past traditional information-theoretic limits. This routing framework can extend to other challenging chemical spaces—such as halide double perovskites ($A_2BB'\mathrm{X}_6$), vacancy-ordered double perovskites ($A_2B\Box\mathrm{X}_6$), and high-entropy oxides—where 3D DFT relaxations are computationally prohibitive. Linking these analytical equations with active-learning Bayesian optimization will enable rapid, autonomous prescreening before committing to full DFT calculations.

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CRediT Authorship Contribution Statement

Nihal Gazi: Conceptualization, Methodology, Software, Formal Analysis, Writing - Original Draft, Data Curation. **Meghneel Ghosh**: Investigation, Software, Validation, Formal Analysis, Writing - Review & Editing. **Subarna Datta**: Resources, Supervision, Validation, Project Administration, Writing - Review & Editing. **Soumyadipta Pal**: Conceptualization, Supervision, Methodology, Project Administration, Writing - Review & Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The datasets curated in this study (the 2,000 baseline double perovskite dataset and the 5,000 topological scaling dataset) along with the complete

Python source code for the Master Physics-Gated Architecture, feature engineering routines, and benchmark evaluation scripts are openly available on GitHub at `https://github.com/nihal-gazi/double_perovskite_compositional_interpretable`.

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